

Tissue-specific interpretation after project, seed, method, and composition review

Thymus

Established context

Thymic involution, reduced proliferation, and adaptive immune suppression.

Retained pathway layer

Lower DNA repair and cytoskeletal regulation, with lower lymphoid-stromal interaction as a model-specific niche hypothesis.

Interpretation boundary: Cell-resolved validation is needed to separate thymocyte state from stromal composition.

Skin

Established context

Barrier injury, inflammation, oxidative stress, and impaired regeneration.

Retained pathway layer

Lower chromatin regulation, DNA repair, Hedgehog, sphingolipid metabolism, and cell-junction organization describe a coordinated maintenance-state decrease.

Interpretation boundary: Protocol depooling shows that gravity, site, recovery, strain, and duration modify the aggregate response.

Liver

Established context

Lipid, xenobiotic, insulin-related, and mitochondrial dysregulation.

Retained pathway layer

Lower MHC class II antigen presentation and T-cell receptor signaling add a reproducible adaptive-immune communication axis.

Interpretation boundary: Bulk data cannot distinguish immune-cell abundance from signaling state.

Spleen

Established context

Reduced T-cell abundance and activation after spaceflight.

Retained pathway layer

Lower T-cell receptor, neutrophil degranulation, and C-type lectin receptor programs connect adaptive suppression with lower innate effector and pathogen-sensing states.

Interpretation boundary: The scores are transcriptomic programs, not direct functional measurements of neutrophils or pathogen response.